

Revised Generation Interconnection System Impact Study Report

For Queue Project Q12-0145

State Campus 138 kV
July 30, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0145. The Interconnection Customer proposes a 1347.7 MW natural gas combined cycle facility interconnecting at the State Campus 138 kV substation in the Pennsylvania area. The request was received on July 25, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 96 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$149,856,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: State Campus Substation-American River 138 KV
Contingency: Kiya-State Campus 138 KV
Loading: 637.0% (Rating: 211 MVA)
DFAX: 100.0%
Pre-project loading: 3.0%
2. Facility: State Campus-Kiya 138 KV Line
Contingency: State Campus-American Riv 138 KV line
Loading: 609.7% (Rating: 220 MVA)
DFAX: 100.0%
Pre-project loading: 2.9%
3. Facility: Glen Ullin Station 6-Kiya 138 KV Line
Contingency: State Campus-American River 138 KV
Loading: 578.2% (Rating: 230 MVA)
DFAX: 100.0%
Pre-project loading: 6.9%
4. Facility: Glen Ullin Station 6-E alamus 138 KV
Contingency: State Campus-American Riv 138 KV L
Loading: 558.9% (Rating: 237 MVA)
DFAX: 100.0%
Pre-project loading: 9.2%
5. Facility: Douglas-Tonopah 138 KV line
Contingency: Kiya-State ampus 138 KV
Loading: 393.2% (Rating: 218 MVA)
DFAX: 66.1%
Pre-project loading: 15.2%

6. Facility: Tonopah-American River 138 KV Line
Contingency: Kiya-State Campus 138 KV
Loading: 378.3% (Rating: 230 MVA)
DFAX: 66.1%
Pre-project loading: 9.8%
7. Facility: E Calamus-Mt Pleasant 138 KV
Contingency: State Campus-American River Substation 138 KV L
Loading: 339.5% (Rating: 215 MVA)
DFAX: 54.7%
Pre-project loading: 3.7%
8. Facility: Longview Power Plant-Mt Pleasant 138 KV
Contingency: American Riv Substation-State Campus 138 KV Line
Loading: 329.2% (Rating: 220 MVA)
DFAX: 54.7%
Pre-project loading: 5.7%
9. Facility: E Calamus-Debary 138 KV Line
Contingency: Amerian Riv-State Campus 138 KV line
Loading: 267.3% (Rating: 221 MVA)
DFAX: 45.3%
Pre-project loading: 9.1%
10. Facility: Debary-Hatfield 138 KV
Contingency: State Campus-American River 138 KV
Loading: 259.3% (Rating: 226 MVA)
DFAX: 45.3%
Pre-project loading: 10.5%
11. Facility: Hatfeld Substation-Cooney 138 KV L
Contingency: State Campus-American Riv 138 KV
Loading: 255.6% (Rating: 228 MVA)
DFAX: 45.3%
Pre-project loading: 12.0%
12. Facility: Longview Power Plant-WestRun 138 KV
Contingency: State Campus-American Riv Substation 138 KV
Loading: 238.3% (Rating: 207 MVA)
DFAX: 36.1%
Pre-project loading: 3.2%
13. Facility: Cooney-Prides Corner 138 KV L
Contingency: State Campus-American River 138 KV
Loading: 228.3% (Rating: 248 MVA)
DFAX: 45.3%
Pre-project loading: 18.2%
14. Facility: Rinn Vly-American River 138 KV L
Contingency: Tonopah-American River 138 KV
Loading: 227.0% (Rating: 210 MVA)
DFAX: 35.6%
Pre-project loading: 1.4%
15. Facility: Matanzs-Payne 138 KV
Contingency: Kiya-State Campus 138 KV
Loading: 221.3% (Rating: 228 MVA)
DFAX: 42.4%
Pre-project loading: 29.5%
16. Facility: Payne-Kress 138 KV
Contingency: Kiya-Sate Campus 138 KV

- Loading: 219.0% (Rating: 232 MVA)
DFAX: 42.4%
Pre-project loading: 27.1%
17. Facility: Douglas-Kress 138 KV line
Contingency: State Campus-Kiya 138 KV
Loading: 214.7% (Rating: 240 MVA)
DFAX: 42.4%
Pre-project loading: 23.7%
18. Facility: Castle Rock-Rin Vly Substation 138 KV
Contingency: Tonopah-American Riv 138 KV
Loading: 207.6% (Rating: 224 MVA)
DFAX: 35.6%
Pre-project loading: 6.0%
19. Facility: Missouri City-Castle Rock 138 KV
Contingency: Tonopah-American River 138 KV
Loading: 202.4% (Rating: 228 MVA)
DFAX: 35.6%
Pre-project loading: 7.8%
20. Facility: West Run-Morgantown Energy Facility 138 KV
Contingency: State Campus-American Riv 138 KV
Loading: 200.5% (Rating: 234 MVA)
DFAX: 36.1%
Pre-project loading: 7.2%
21. Facility: Missouri City-Wolf Road 138 KV
Contingency: Tonopah-American Riv 138 KV Line
Loading: 189.8% (Rating: 239 MVA)
DFAX: 35.6%
Pre-project loading: 11.0%
22. Facility: Vancouver Island-Wolf Rd 138 KV line
Contingency: Tonopah-American River 138 KV
Loading: 178.7% (Rating: 248 MVA)
DFAX: 35.6%
Pre-project loading: 14.5%
23. Facility: Cwec Iola-Vancouver Island 138 KV Line
Contingency: Tonopah-American Riv 138 KV L
Loading: 169.8% (Rating: 255 MVA)
DFAX: 35.6%
Pre-project loading: 17.9%
24. Facility: Douglas-Anchorage 1 138 KV
Contingency: State Campus-Kiya 138 KV
Loading: 145.8% (Rating: 227 MVA)
DFAX: 23.7%
Pre-project loading: 5.3%
25. Facility: Anchorage 1-Griffin 138 KV Line
Contingency: State Campus-Kiya 138 KV line
Loading: 142.3% (Rating: 230 MVA)
DFAX: 23.7%
Pre-project loading: 3.5%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
State Campus Substation-American River 138 KV	Rebuild line with higher-capacity conductor	\$9,510,000	100% (\$9,510,000)
State Campus-Kiya 138 KV Line	Rebuild line with higher-capacity conductor	\$5,208,000	100% (\$5,208,000)
Kiya-Glen Ullin Station 6 138 KV	Rebuild line with higher-capacity conductor	\$2,135,000	100% (\$2,135,000)
Glen Ullin Station 6-E alamus 138 KV	Rebuild line with higher-capacity conductor	\$4,589,000	100% (\$4,589,000)
Douglas-Tonopah 138 KV	Rebuild line with higher-capacity conductor	\$5,807,000	100% (\$5,807,000)
Tonopah-American Riv 138 KV	Rebuild line with higher-capacity conductor	\$4,457,000	100% (\$4,457,000)
E Calamus-Mt Pleasant 138 KV	Rebuild line with higher-capacity conductor	\$14,273,000	100% (\$14,273,000)
Mt Pleasant-Longview Power Plant 138 KV	Rebuild line with higher-capacity conductor	\$5,996,000	100% (\$5,996,000)
Debary-E Calamus 138 KV	Rebuild line with higher-capacity conductor	\$2,950,000	100% (\$2,950,000)
Debary-Hatfield 138 KV line	Rebuild line with higher-capacity conductor	\$2,799,000	100% (\$2,799,000)
Hatfeld Substation-Cooney 138 KV L	Rebuild line with higher-capacity conductor	\$18,284,000	100% (\$18,284,000)
W Run-Longview Power Plant 138 KV	Rebuild line with higher-capacity conductor	\$6,104,000	100% (\$6,104,000)
Cooney-Prides Corner 138 KV	Rebuild line with higher-capacity conductor	\$2,716,000	100% (\$2,716,000)
Rinn Vly-American River 138 KV L	Rebuild line with higher-capacity conductor	\$2,890,000	100% (\$2,890,000)
Matanzas-Payne 138 KV	Rebuild line with higher-capacity conductor	\$10,511,000	97% (\$10,213,000)
Kress-Payne 138 KV	Rebuild line with higher-capacity conductor	\$2,972,000	97% (\$2,888,000)
Douglas-Kress 138 KV line	Rebuild line with higher-capacity conductor	\$3,714,000	97% (\$3,609,000)
Castle Rock Substation-Rinn Vly 138 KV	Rebuild line with higher-capacity conductor	\$5,042,000	100% (\$5,042,000)
Castle Rock-Missouri City 138 KV	Rebuild line with higher-capacity conductor	\$18,215,000	100% (\$18,215,000)
West Run-Morgantown Energ Facility 138 KV	Rebuild line with higher-capacity conductor	\$2,403,000	100% (\$2,403,000)
Missouri City-Wolf Road 138 KV	Rebuild line with higher-capacity conductor	\$7,220,000	100% (\$7,220,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Vancouver Island-Wolf Rd 138 KV	Rebuild line with higher-capacity conductor	\$1,883,000	100% (\$1,883,000)
Cwec Iola-Vancouver Island 138 KV Line	Rebuild line with higher-capacity conductor	\$1,899,000	100% (\$1,899,000)
Douglas-Anchorage 1 138 KV	Rebuild line with higher-capacity conductor	\$3,823,000	97% (\$3,710,000)
Anchorage 1-Griffin 138 KV Line	Rebuild line with higher-capacity conductor	\$5,210,000	97% (\$5,056,000)
	Total allocated to project		\$149,856,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.